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VoIP Sentinel S11

User's Manual

VoIP Sentinel Doorstation Phone



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1.0 Introduction

Thanks you for choosing Dallas Delta products. We hope that our product will meet the demands of your application needs and it will give you many years of trouble free use.

The **VoIP Sentinel** is an ethernet connected telephone that provides voice over internet Protocol (VoIP) communication technology.

Giving you the power of our PSTN base units in a IP format. Using a SIP protocol standard, it provides an easy connection to most VoIP based equipment.

The Sentinel is a loud speaker, hands-free unit. It is well suited for homes, hotels, hospitals and universities as well as many business and commercial settings and providing remote gate access guarding you against unwanted entry onto your property.

This document outlines the units Features, Operation, Programming instructions and Installation procedures.



2.0 How to Use this Manual

This manual is set out with the intended use of both installers and end users in mind. It will give you the ease and quick turn-around time of installation required to get the job done efficiently and will also give you a very detailed understanding of the unit you are installing and putting into real-world use.

It is very important that installation of this unit is carried out by a qualified personnel to avoid damage of the unit and correct functionality of the product. It will also ensure that safely requirements are met for everyday use of this electrical product.

When installing this product for the first time, it is critical to follow the quick start guide to begin with. This will ensure that your new product is up and running in a very simple and easy manner with as minimal time as possible.

It is then recommended that the manual be read from cover to cover to allow you to fully understand the capabilities of your product and be able to obtain the most of this unit.



3.0 Quick Start Guide

3.1 Getting Started - Basic Procedures

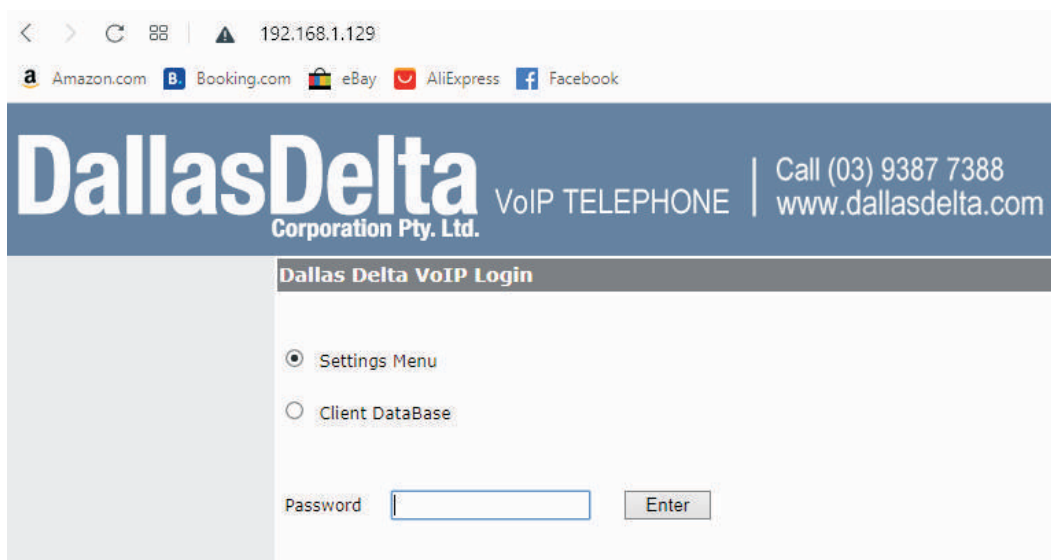
There are a few basic procedures that will need to be completed in order to get your VoIP Doorstation working immediately. Please follow these simple procedures in order before attempting to use your phone:

- > Log on to your VoIP Doorstation
- > Enter basic 'SIP Proxy' parameters
- > Enter basic 'Systems' parameters
- > Enter basic 'Network' parameters
- > Program your customer database numbers for each client.

Below are very simple to follow step-by-step procedures to getting your VoIP Doorstation up and running:

3.2 Logging On to the Home Page

In order to begin setting up your VoIP Doorstation you must first log into the unit via a web browser (such as Google Chrome, Firefox, MS Edge, etc.). log into your VoIP doorstation by typing in the IP address of the phone in the address bar.



Note: No password is programmed when dispatched from factory setup.

To obtain the IP address from a unit enter *****#7*0** and the address will be shown on the LCD.

The phone can be set to DHCP or Static LAN.

To configure for DHCP, enter *****#7*2**
or to set the unit to STATIC IP enter *****#7*1** This will set the phone to 192.168.1.100.



3.3 Web page Access

Sentinel web pages are in 2 sections. The first is for setting up the phone VoIP registration and functional parameters necessary for the operation of the phone.

The 2nd section is the database for each client (up to 1024). Requiring Room numbers, Phone number and Relay codes for gate access.

To access the phone via the Ethernet will require the phone's IP address.

This can be done by following the key sequence as shown below.

When retrieving the IP address, it can either return a IP if set to Static or Dynamic mode or it can return 0.0.0.0 if set to DHCP and there is no DHCP server on the LAN.

If it does return 0.0.0.0, then set the phone up for Static IP step 3.3.2

3.3.1 Retrieving the IP Address

To check the IP address of the phone,

Enter *** # * # 7 * 0**

on the keypad during standby mode and it will play back a voice message with the current IP address.

3.3.2 Reset Static IP Address

This will set the address to 192.168.1.100

To set the phone to Static IP address, use the following key sequence:

*** # * # 7 * 1**

3.3.3 Reset DHCP IP Address Range

To set the phone's IP address to the DHCP range, use the following key sequence:

*** # * # 7 * 2**

3.3.4 Reset the phone back to factory defaults

To reset the phone's parameter back to a clean start (factory reset). Enter the following key sequence:

*** # * # 7 * 9**

Note: This will reset the IP address to static. 192.168.1.100
VLAN to 0.

Clear the Admin PIN. and

Erase the all SIP proxy details.

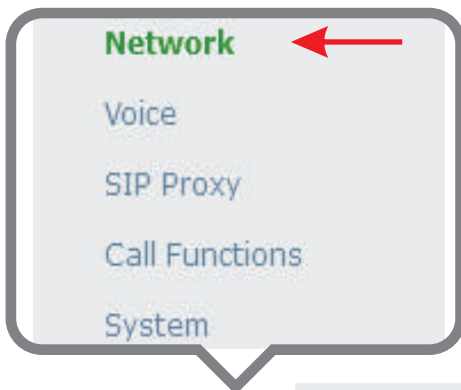


3.0 Quick Start Guide

3.4 Settings

3.4.1 Network Settings

Note changing the network settings at this stage will required you to re-log back into the phone if the IP address is changed. If so, follow the steps outlined in 3.2 *Logging on to the Home Page*.



This page will show the current status of the registration, MAC ID, and version number of the F/W.

If your phone does require a static IP location, then verify with the IP department which Address can be used. Else leave the phone in DHCP mode to obtain an IP automatically.

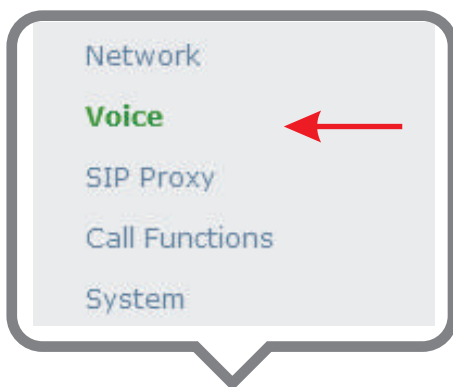
If DHCP is selected, then most likely to leave the DNS to AUTO. Care should be taken when setting the VLAN data, Refer to the IP department for these settings.

The Status is updated every 8 seconds.

Phone Status	
Phone Model:	DDC_VoIP-Sent
MAC Address:	38:b1:9e:d0:00:c8
Version No:	s11.0420.23
SIP Registered :	Yes > Server 1
SiteTemperature :	~ 17 (DegC)

Network Settings	
Connection Type:	DHCP
IP Address:	192.168.1.100
Subnet Mask:	255.255.255.0
Default Gateway:	192.168.1.1
DNS Server IP:	Auto DNS
Primary DNS:	192.168.1.1
Secondary DNS:	
Layer 3 QoS:	0 (DSCP Diff-Serv or Precedence value)
Layer 2 QoS:	0 802.1Q VLAN Tag
	0 802.1p Priority Value

OK Cancel



3.4.2 Voice Codec

Most PBX phone systems today will work with the codec set to G711 (PCMA or PCMU). Select preference and the order as required.

Note: The PBX can over-ride this settings to match the other party as part of the SIP protocol. If a match can not be made the call will terminate.

The RTP voice frame is in 10mS segments and sent with a network packet. How many frames are sent with each packet is set by the 'Frames/Package' option.

For most cases you would normally send 2-3 frames/package.

Voice Codec Settings				
Preferred Voice Codec:	Codec 1:	PCMA	Frames/Package 1:	2
(In listed order)	Codec 2:	PCMU	Frames/Package 2:	2
	Codec 3:	G.726-32	Frames/Package 3:	2
	Codec 4:	G722	Frames/Package 4:	2
	Codec 5:	G.729	Frames/Package 5:	2



3.0 Quick Start Guide

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Voice

SIP Proxy

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3.4.3 SIP Proxy Settings

Select 'SIP Proxy' on the side menu to open up the SIP Proxy settings page.

To setup a basic SIP Registration you will need details setting from the site SIP PBX.

The PBX needs to be configured for a *3rd party SIP profile*.

Then fill in the minimum required fields to complete the registration as follows:

Critical items that are required for SIP registrations are:-

SIP Server: & SIP Domain: If the Outbound Proxy has a different address to the Domain, then set the Outbound Proxy in the SIP Server field, and enable the '**Use the SIP Server As Outbound Proxy:**' option. and

SIP User ID_Name: , SIP Authentication ID: & SIP Authentication PIN: (Note, This password is sometimes refer to as Digest KEY).

Other options need to be checked with the IT department.

Complete the necessary fields then click "OK" to save. Then return to the Network Page and view the "Phone Status" to confirm if the phone does register correctly.

SIP Proxy Settings	
SIP Server 1:	192.168.1.90 (IP or URI)
SIP Domain 1:	192.168.1.90
SIP Server 2:	
SIP Domain 2:	
SIP Server Port:	5060 (Typically 5060)
SIP Registration:	☒
Use the SIP Server As Outbound Proxy:	☐
Get SIP details from the DNS SRV:	☐
Support PRACK:	☐ (100rel, RFC3262 protocol)
SIP User ID_Name:	906
SIP Authentication ID:	906
SIP Authentication PIN:	 (Digest Key)
Displayed Name:	Front Gate (optional, e.g., Gate #1)
Local SIP Port:	5060 (Typically 5060)
Local RTP Port:	6000 min 10000 max (65534)
Register Expiration:	600 (In seconds, Typically 120s)
Keep Alive Interval:	20 (In seconds, Typically 20s)
Registration Attempts:	5 (before phone restarts, Typically 5)

VoIP Doorstation SIP Proxy Settings Page



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3.4.4 Call Functions

In this section select what should occur for various in-coming or out-going calls. Generally, tenants would not be calling the Sentinel. This should only be done by a selected few.

If the option '*Enable White List*' is ticked then only the numbers listed in the '**Phone Number**' table can call the Sentinel.

If It is required to divert all calls to a central location, IE the caretaker, then set the '*Divert numbers*' and the '*Timeout*' as required.

The call forwarding conditions will call the '*Forward-to Number*' for the event set by *Unconditionally, Busy or No-Answer*. If none of those events are set and the '*No Answer period*' is set, then the phone will Auto-answer if set or terminate the in-coming call.

The '*Dial Prefix*' is only used for unit that have Call buttons installed.

Calling Functions	
Forward-to Number:	(For Incoming Calls)
Forward Unconditionally:	☐
Forward When Busy:	☐
Forward When No Answer:	☐
Incoming, No Answer Period:	5 (In seconds)
Auto Answer:	☒
Enable White List:	☒ Only allow incoming calls from the "Phone Number" table.
Dial Prefix:	(Prefix digits added to Button numbers)
Hot Line Number:	(Auto-dialed number on handset pickup)
Divert Number 1:	(For Outgoing Calls, on Busy or No Answer)
Divert Number 2:	
Calling, No Answer Period:	0 (In seconds)
Divert for buttons only:	☐ (Only divert calls when a button is pressed)

VoIP Doorstation Call Functions Page



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Configuration

KeyLess Codes

3.4.5 System Settings

In this section you should set the Administration Password used to access these web pages.

It is very important to document this PIN with the site administrator.

For units that are installed with a Handset, set the option to on and the volume levels as required.

For 'loud-speaker' type phones set the volume levels as best suited to the site.

The 'conversation period' will terminate the call when it expires. This is set to limit the call duration.

Each tenant will have their own codes to access/activate the relays. If the setup prefers to use the same relay code for all tenants then set these code here and enable the global relay option switch.

When a call is made to the phone or if a call is diverted, then the person here would use the 'in-coming & divert' relay codes to grant access.

The relay on-time is used for all relay activations.

The Sentinel can record all events made to a Sys-log Server, One should be set-up for security. The Syslog-IP should be set to this sever.

The Time of day that maybe displayed on the LCD is set by the NTP server. If the phone does not have access to the internet then set this server to the SIP server.

System Settings		
Administration Password:		(Password to enter this web)
Handset Installed:	☐	
Handset input (0-30):		ACA level = 15 to 20
Handset Output (0-30):		ACA level = 20 to 30
Microphone input (0-30):		15
Speaker Output (0-30):		15
Ring Volume (10-30):		15
Dial tone Vol (0-30):		15
Conversation Timer (0-99):		3 (In minutes)
Use Global relay codes	Relay1:	Relay2: ☐ Enable Sw
Code for In-coming and diverted calls	Relay1:	Relay2:
Relay On Timer (0-30):		5 (In seconds, Typically 5s)
Syslog IP:		255.255.255.255
Syslog time Intervals:		0 (In minutes, 0=off, max-65535 min)
Enable Debug output:	☒	
NTP Server:		0.au.pool.ntp.org (IP or URI) eg, 0.au.pool.ntp.org
Time Zone:		(GMT+10:00)Canberra,Melbourne,Sydney
Adjust Clock for Daylight Saving:	☒	

VoIP Doorstation System Settings Page



3.0 Quick Start Guide

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LCD Messages

3.4.6 Configuration

If the on board relays are not used to grant access to 2 gates. Then setting one relay to activate whenever the phone is being used for a purpose of lighting or a camera can be achieved by selecting the 'Relay Inuse' option.

The option 'Dial Number Silently' will mute the digits dialled when calling the tenant number.

If it is preferred to end the call after a relay code is accepted then enable the 'Disconnect After Gate open' option.

The SNMP server maybe used to log events from the phone (this is an alternative to the Sys-log server), The IP address, here will collect trap events. It uses version 2c SMNPtraps which requires a 'Community Name'.

Phone Configuration

Relay Inuse	None
Dial Numbers Silently:	☒
Disconnect After Gate open:	☐
SNMPv2c Trap Destination_1:	0.0.0.0 (SNMPv2c traps server IP address)
SNMPv2c Trap Destination_2:	0.0.0.0
SNMPv2c Community Name:	DDCVoIP
Auto Provisioning Server:	(IP or URI)
Auto Provisioning Port:	(Typically 80)
Auto Provisioning Interval:	(In minutes, 60 to 65535 minutes)

Hardware Settings

Relay1 Default State	Relay N/O	Note, When power is off, Relay N/C state =open
Lift Controller Unit Installed	☐	(IP) (On Time, seconds)
Temperature calibration:		(Offset +/-10)



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3.4.7 Keyless Codes

Emergency services or site administrators can use the keyless system to gain access. They will need to enter a * plus the one of the codes. This will activate either relay for the matching group.

For security these codes should be 6 or more digits long. The LCD will indicate 'Welcome' when there is a match.

Keyless Codes			
Relay 1 group:	123		
	456		
Relay 2 group:	789		
	987		

Keyless Access Codes for Emergency Services or Tenants.
To gain entry at the gate, user will type * plus one of the above codes

KeyLess Codes

LCD Messages

Phone Numbers

3.4.8 LCD Messages

The 2 line LCD display on the Sentinel may be set to display either the time of day or a fixed 16 character message. Alternatively scroll a 63 characters long message.

If line 1 is blank and not used for a scrolling message, then the time of day is displayed here.

Scrolling messages display rate is set by the speed option.

LCD Messages	
LCD Controls	LCD Scroll Speed
Scroll Line 2	4 (0 to 7)
Top Line Scroll msg	
Bottom Line Scroll msg	
Enter the room number then press # to call.	
Top Line Fixed msg	
102 Albert St.	
Bottom Line Fixed msg	



3.0 Quick Start Guide

KeyLess Codes

LCD Messages

Phone Numbers

3.4.9 Phone Number

For Sentinels that are also fitted with buttons then the number called is set within this section.

Note, that prefix digits that are set in section 3.4.4 are added to the beginning of these number.

The numbers in this table can also be used to restrict who can call the phone.

This is enabled by setting the 'Enable White List' option and add these numbers here.

Button input	Name	Phone Number
001	Button1	1010
002		
003		
004		
005		
006		
007		
008		
009		
010		
011		
012		
013		
014		
015		
016		

Push Button phone numbers or Numbers to be use in 'White List'



4.0 Programming Instructions

Network Settings

Phone Status will be display Model, MAC address and the SIP Registering status, (this is updated every 8 seconds).

And the submenu's are shown on the left of the WEB page.

The 'Network settings' will enable you to change IP address, subnet mask and gateway and the DNS server.

Network Voice SIP Proxy Call Functions System Configuration KeyLess Codes LCD Messages Phone Numbers Contact Us	Phone Status	
	Phone Model: DDC_VoIP-Sent	
	MAC Address: 38:b1:9e:d0:00:c8	
	Version No: s11.0420.23	
	SIP Registered : Yes > Server 1	
	SiteTemperature : ~ 24 (DegC)	
	Network Settings	
	Connection Type:	DHCP
	IP Address:	192.168.1.100
	Subnet Mask:	255.255.255.0
	Default Gateway:	192.168.1.1
	DNS Server IP:	Auto DNS
	Primary DNS:	192.168.1.1
	Secondary DNS:	
	Layer 3 QoS:	0 (DSCP Diff-Serv or Precedence value)
Layer 2 QoS:	0 802.1Q VLAN Tag	
	0 802.1p Priority Value	

Use the **Connection Type** to set either *STATIC IP* or *DHCP* address for the phone.

If the connection is DHCP then the IP address, Mask and Gateway are automatically assigned by the network DHCP server.

When set to a Static IP, Set the IP address, Subnet Mask and Gateway address that will suit the local area network (LAN).

If you change the IP address you are then required to restart by logging in using the new IP address.

If the site does have a DHCP sever the 'DNS Server IP' can be set to **Auto DNS**. In this mode the DHCP sever will assign the DNS automatically.

If not, enable **Manual DNS** and set the Primary and Secondary DNS address.

If the network switch provides QoS traffic control, then set the **Layer 2 & 3 QoS** values to match, else leave these values to '0'.

For Layer 3 QoS, a value in the field of 46 equals *EF (Expedited Forwarding)*.

Set the VLAN tag option for networks with VLAN option enabled.

Take care here as once this is set then the PC/Laptop you are using may not have the VLAN option enabled. You will not be able to communicate with the phone.



4.0 Programming Instructions

Voice

In this page, select the order and preferred codecs, plus the amount of frames per packet to sent for each codec type.

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Voice Codec Settings

Preferred Voice Codec: (In listed order)

Codec 1:	PCMA ▼	Frames/Packet 1:	2
Codec 2:	PCMU ▼	Frames/Packet 2:	2
Codec 3:	G.726-32 ▼	Frames/Packet 3:	2
Codec 4:	G.722 ▼	Frames/Packet 4:	2
Codec 5:	G.729 ▼	Frames/Packet 5:	2

Frames/Packet Range: Typically 1 to 4

Send DTMF: RFC2833 ▼

DTMF RFC2833 Payload: 101 (Between 96 and 127, Typically 101)

G.726-32 Payload Type: 111 (Between 96 and 127, Typically 111)

When selecting the codecs consider the data rate that will be required by your network. As a guild the list here show the data bit rate required for each codec, if the network is via a low speed ADSL then the PCMu/a may not be a suitable choice.

Table 1. Codec comparison

Codec	bit rate (kbit/s)
G.711(PCMu /a)	64
G726-32	32
G.729	8
G.722	32

The **Frames/Packet** selects how many blocks of voice data (10mS segments) are transmitted with each packet sent over the network, typically 2 - 4 is recommended.

The **Send DTMF** selects how the unit will send **DTMF** digits over the network. There are 2 protocols available, RFC2833 and INFO.

The remaining options set the payload required for each protocol and it is recommend that the default values be used.



4.0 Programming Instructions

SIP Proxy

The parameters on this page will identify the SIP server and SIP authentication details. Your ISP or network administrator will provide the necessary information to be used here.

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www.dallasdelta.com

VoIP TELEPHONE

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SIP Proxy Settings

SIP Server 1:	192.168.1.90	(IP or URI)
SIP Domain 1:	192.168.1.90	
SIP Server 2:		
SIP Domain 2:		
SIP Server Port:	5060	(Typically 5060)
SIP Registration:	☒	
Use the SIP Server As Outbound Proxy:	☐	
Get SIP details from the DNS SRV:	☐	
Support PRACK:	☐ (100rel, RFC3262 protocol)	
SIP User ID_Name:	909	
SIP Authentication ID:	909	
SIP Authentication PIN:	•••••	(Digest Key)
Displayed Name:	Front Gate	(optional, e.g., Gate #1)
Local SIP Port:	5060	(Typically 5060)
Local RTP Port:	6000 min 9000 max	(65534)
Register Expiration:	600	(In seconds, Typically 3600s)
Keep Alive Interval:	20	(In seconds, Typically 20s)
Registration Attempts:	5	(before phone restarts, Typically 5)
Proxy Require:		
NAT traversal:	Disabled	
NAT IP:		
STUN Server:	(IP or URI)	
STUN Server Port:	(Typically 3478)	

SIP Server1 will be the main running location and the **SIP Server2** can be the fail over location if available. The User detail will be the same in both servers

When **SIP registration** is required with the server then enable this option. If the phone is configured to dial a direct IP location and no SIP server is used then leave this option disabled.

If the server is via an Outbound Proxy, then use the **SIP Server** field for this proxy and enable the option **Use SIP Server as the Outbound Proxy**.

The user ID's, and PIN are supplied by the ISP/Network administrator, this information is case sensitive and should be typed in correctly. The **Displayed Name** is relayed to the connecting IP telephone as a caller ID and is optional. (Not all PBX relay this field).

SIP and **RTP** port numbers are sent to the router/firewall and are use to provide paths for voice and control data to and from the phone.

The **Register Expiration** and the **Keep Alive** timers keep the SIP server mapping and NAT ports open. The **Registration Attempts** sets how many failed attempts before rebooting the phone, this maybe preferred to re- obtain a new IP from the DHCP server and start from fresh.

If a *Provisional Response Acknowledge PRACK* is required within your network, then enable the **Support PRACK(100rel,RFC3262)** option.

When network server is behind a NAT, enable the **NAT Traversal** and set either the **NAT IP** or **STUN server** address and port as required.



4.0 Programming Instructions

Calling Functions

This section controls the way the unit makes and receives calls.

Calling Functions	
Forward-to Number:	(For Incoming Calls)
Forward Unconditionally:	☐
Forward When Busy:	☐
Forward When No Answer:	☐
Incoming, No Answer Period:	5 (In seconds)
Auto Answer:	☒
Enable White List:	☒ Only allow incoming calls from the "Phone Number" table.
Dial Prefix:	(Prefix digits added to Button numbers)
Hot Line Number:	(Auto-dialed number on handset pickup)
Divert Number 1:	(For Outgoing Calls, on Busy or No Answer)
Divert Number 2:	
Calling, No Answer Period:	30 (In seconds)
Divert for buttons only:	☐ (Only divert calls when a button is pressed)

Forward to Number will set the telephone number that will be called when calls are redirected by one of three methods:

- 1) if the **Forward unconditionally** is set to yes (any calls made to the phone will be immediately redirected).
- 2) **When on busy** is set, then any attempt to call the unit when the phone is in use will be redirected.
- 3) Any un-answered call will also be redirected to this number when the **Forward When No Answer** switch is set and the **Incoming, No Answer Timeout** period has elapsed. (*The Auto-answer option needs to be disable, for this to function*).

The **Incoming, No Answer Timeout** period will set how long the phone will ring for.

If the **Auto Answer** switch is disabled then the in-coming call is terminated after that period.

An option to restrict in-coming calls to a limited list can be enabled by selecting the **Enable White List**

This will only allow calls from the numbers within the phone book. You may add these numbers in any location within the **Phone Number** section.

The digits in the **Dial Prefix** are added to the any number dialled via the preprogrammed button numbers.

For DDC_VoIP phones that have a handset, a hotline number (dialled when the handset is pickup), maybe programmed in the field **Hot Line Number** field.

When the DDC_VoIP phone is making a call, it is possible to divert the call to 2 alternative numbers if the other party is busy or doesn't answer.

To enable this function, enter the **Divert Numbers** and set the **Outgoing, No Answer Period** a value in second.

The **Divert for buttons only** is used if your unit has both buttons and a keypad installed.



4.0 Programming Instructions

System Settings

The DDC_VoIP phone audio levels, relay options, Syslog details and time of day functions is set in this section. Audio levels for the phone should be re-check on site.

The on board relay/s may be use for gate access control, back ground lighting / camera. They can be remotely activated or enable when the phone is in use.

System Settings	
Administration Password:	(Password to enter this web)
Handset Installed:	☐
Handset input (0-30):	ACA level = 15 to 20
Handset Output (0-30):	ACA level = 20 to 30
Microphone input (0-30):	15
Speaker Output (0-30):	15
Ring Volume (10-30):	15
Dial tone Vol (0-30):	15
Conversation Timer (0-99):	3 (In minutes)
Use Global relay codes	Relay1: 5599 Relay2: 9955 ☐ Enable Sw
Code for In-coming and diverted calls	Relay1: 999 Relay2: 888
Relay On Timer (0-30):	5 (In seconds, Typically 5s)
Syslog IP:	255.255.255.255
Syslog time Intervals:	0 (In minutes, 0=off, max-65535 min)
Enable Debug output:	☒
SNTP Server:	0.au.pool.ntp.org (IP or URI) eg, 0.au.pool.ntp.org
Time Zone:	(GMT+10:00)Canberra,Melbourne,Sydney ▾
Adjust Clock for Daylight Saving:	☒

The **Administration Password** is used to gain access to the web page. For added security set and document this password. The phone is supplied with no password set.

The microphone and the speaker levels for the handset (if fitted) and in the hands-free circuit, is adjusted to suit site background noise, when possible keep these level low, as un-necessary noise will be transmitted back to the remote operator.

If the phone does have a handset installed, then enable the option and set the audio levels as required for the handset.

NOTE: ACA maximum levels:

Do not exceed the handset maximum levels, for conformity and comfort keep handset levels to a minimum.

Set the **Ringer Volume** and **Dial Tone** levels to suit site conditions.

Conversation period may be limited if desired, to set this, enter a period in minutes into the **Conversation Timer** field. If no timer is required then enter a 0 to disable this function.

The onboard relay may be activated remotely when the remote operator dials the digits that matches the codes for **Relay 1** & **Relay 2**.

'**Global Relay Codes**' if enabled will be used by all tenants, these codes will override the codes set in the data-base. But only for the calls they receive. Client codes are till used for keyless access.



4.0 Programming Instructions

System Settings continue.

Use the **Relay on timer** field for the duration required. Or set it to 0, to remain on for the duration of the call.

Remote Relay Module

Remote Relay option (if installed) will be use to active a relay via the LAN ethernet. The remote relay device can activate 1 of 3 relays. If required set the IP address of the '**Remote relay controller**' and the relay '**code**'. This code PLUS a relay number is used by the operator to trigger the relay. IE, If the code is 5, then to set the relay, the operator would enter 5 plus the digit 1-3 to active relay 1 to 3. *Note that the Remote Relay Controller may not be installed with all the relays.* Also please refer to the manual supplied with the relay module.

The remote relay module may also be used to activate a relay whenever the phone is in-use. If this is required select this option and the relay to active.

Debug Options:

Debugging can be enabled by **Syslog** and/or broadcast UDP. If this is not required then disable these options.

For some installation, it may be required to record events that occur at the phone. These events may include;

- which button is pressed
- when a relay is activated remotely
- when the handset is on or off hook
- on ringing (incoming call)
- system reboot
- start of call and at auto answer
- at the end of a call
- and a status of it's current state, ie Logged-on or not at set intervals.

This information is stored at an allocated server and the protocol used is **SYSLOG on port 514**. Although generally on a Linux systems, **SYSLOG** programmes can be sought for the Windows system as well.

Each event is timed stamp with the phone ID and IP address. To enable this function set the IP of the **SYSLOG** server and the intervals.

The DDC_VoIP phone outputs general debugging information on a continuous basis on port UDP 8225. If high traffic is a concern then disable the **Enable Debug output** option.

The 'time of day' used within the phone and for Syslog is set either from the SNTP server or the SIP server.

SNTP Time Server

The **SNTP Server** (simple network time protocol) provides 'time of day' for syslog events. SNTP is on port '123'. If the phone is connected to the internet then a site like 'time.windows.com' or '0.au.pool.ntp.org' may be used and in many cases the SIP Server.

Select your the State in the **Time Zone** option and whether to use '**DayLight saving**' time adjust in the next 2 fields.

If the LAN does not have access to a SNTP server then set this field to 0.0.0.0



4.0 Programming Instructions

Configuration

This configuration section control some of the hardware and functions of the phone.

The '**Relay inuse**' will allow either relay to switch on whenever the phone is being used. It will set the relay on when making and receiving a calls and stay on for the duration of the call.

When a call is made by pressing the button or via the Speed-Dial key, the tones of each digit may be muted, very useful for long phone number. If this suits, then enable the '**Dial Number Silently**' option.

Phone Configuration	
Relay Inuse	None v
Dial Numbers Silently:	☒
Disconnect After Gate open:	☐
SNMPv2c Trap Destination_1:	0.0.0.0 (SNMPv2c traps server IP address)
SNMPv2c Trap Destination_2:	0.0.0.0
SNMPv2c Community Name:	DDCVoIP
Auto Provisioning Server:	(IP or URI)
Auto Provisioning Port:	(Typically 80)
Auto Provisioning Interval:	(In minutes, 60 to 65535 minutes)
Hardware Settings	
Relay1 Default State	Relay N/O v Note, When power is off, Relay N/C state =open
Lift Controller Unit Installed	☐ (IP) (On Time, seconds)
Temperature calibration:	(Offset +/-10)

Set the option '**Disconnect after gate open**' if needed.

Generally, after the relay has been activate remotely there is no further reason to continue in conversation, the phone can be set to terminate the call automatically by enabling this option.

In addition to the other debugging options, this product also can send SNMP traps to a server.

It uses v2c protocol, This protocol requires your community name as well as one or two servers IP addresses. It has an enterprises code of **1.3.6.1.4.1.45255.1.1.1.0 to .10**.

A typical Get command would be '**snmpget -v2c -c DDCVoIP 192.168.1.100 1.3.6.1.4.1.45255.1.1.1.0**'.

The **Auto Provisioning Server** maybe set up if the company has many phones on the network, allowing the admin officer to make changes to the phone as an alternative method to accessing them via the web page. Generally, for few phones the web access is an easier method.

Auto-provisioning sets the parameter in the settings menu and does not set any options in the Client database.

Hardware Settings

Setting the **Relay1 Default State** to N/C (normally closed) will provide the ability to keep the access door closed, but in case there is a power failure to the phone the relay will open.

The Temperature output seen on the Status section can be offset by adding to the '**Temperature Calibration**' value. Note that this value is only an approximation of +/-5 DegC, and only to be used as an indication of the site temperature.



4.0 Programming Instructions

Keyless Codes

Keyless Codes			
Relay 1 group:	123		
	456		
Relay 2 group:	789		
	987		

Keyless Access Codes for Emergency Services or Tenants.
To gain entry at the gate, user will type * plus one of the above codes

Keyless Entry Codes.

These codes would not normally be used by tenants. The tenant should use their own code for security.

Its function is mainly for emergency or site administrators to gain access.

This will activate the relay if there is a match for that group. The same code can not be used in both groups.

The period that the relay stays on for is the same as that set in the relay option section.

Ensure that this period is set.

The user is required to enter * and the code to open the gate.

For sites that require an Emergency Services access, then it may be advisable to use a code that is 8 to 10 digits long. As it may not be changed that easily once the code is supplied to the services personal.



4.0 Programming Instructions

Phone Numbers

The DDC_VoIP phone can be configured with up to 16 buttons,

Each Button, (if fitted) may be set to dial a different phone number. The Name field giving to each input is for internal reference only and is not required.

The phone number may be an IP address/port number, ie 192.168.1.123:5060. If the unit is not registered to a SIP server.

Each number maybe set to a URI location like '912345678@mysipproxy.org'

The phone numbers within this list maybe used to restrict who can call the DDC_VoIP phone. Refer to the 'Enable White-list' option in the **Call Function** section.

	Button input	Name	Phone Number
	001	Care Taker	0415123456
	002	Office	93877388
Network	003		
Voice	004		
SIP Proxy	005		
Call Functions	006		
System	007		
Configuration	008		
KeyLess Codes	009		
LCD Messages	010		
Phone Numbers	011		
	012		
	013		
Contact Us	014		
	015		
	016		

Special commands within the phone number,

The DDC_VoIP unit can be configured to connect to a PSTN/PBX via a ATA. In this scenario the unit calls the ATA and then, on answer, the phone then dials the PBX number.

To do this, enter the number or the IP address of the ATA, followed by a pipe character '|', and then PSTN telephone number to call. (the PBX number may include a pause by adding a comma ','). Example, if the ATA number is 1234 and the PBX number is 9,91234567, then you would enter 1234|9,91234567.

Note, if all Button calls are made via the ATA, then the number can be loaded in the **Dial Prefix** location, In this case the Dial Prefix number would be '**1234|**'.

It may be required to dial this number without each DTMF tone played out over the speaker, especially so, for long phone numbers like an IP address, to mute dialling, enable the option **Dial numbers silently**.

5.0 Client DataBase

The DDC Sentinel VoIP phone can store up to 1024 tenants.

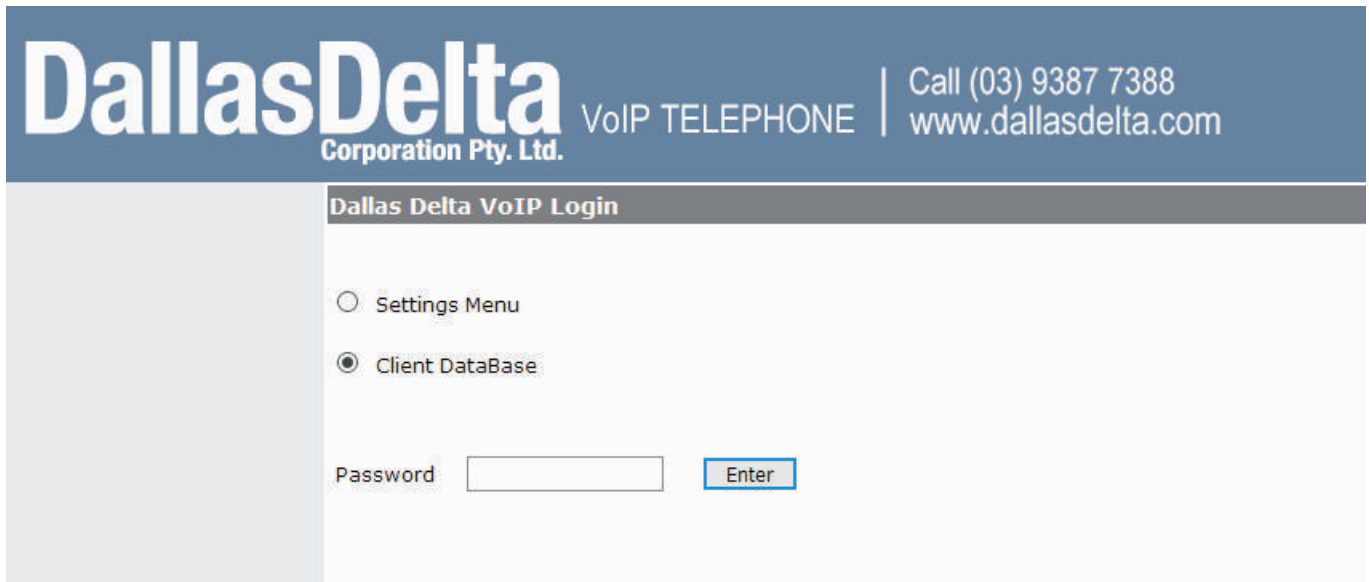
Calls may be established in 2 ways;

- a) by scrolling through the names via the UP - DOWN arrow buttons, or,
- b) by entering the **Room Number** followed by a # key.

Each room number field can be up to 6 digits, and a telephone number field of 32 digits.

The client has access to both relays, each 4 digits long.

The Name field is displayed on the LCD (16 character max).



To edit the Clients data, login to the Sentinel's web page as described earlier in the manual and select '**Client DataBase**' from the login page.

Enter the password if it was previously set, then click 'Enter'.

The data base is shown in groups of 128 records, there are 8 table to make the capacity of 1024 records.

Used the navigation keys at the top of each table << or >>, to display all records.

To the left of the tables there are 2 buttons to access data from PC storage and 2 to access data from the phone storage.

Before any writes to the phone, it will be required to save the data to the PC first.

Ensure that All data is saved prior to clicking **Exit** or **Clear All**.

The file is saved to the PC storage as a CSV.

Be careful when access this data from a common spread sheet program as they will usually tend to modified the format of the data. This may make it unsuitable to re-load back into the web page.

WARNING

This telephone can not be used for emergency purposes during power failure unless fitted with a backup 12 volt battery AND network connection guaranteed.

To be installed and maintained by authorised service personnel only.

DallasDelta

Corporation Pty. Ltd.

VoIP TELEPHONE

Call (03) 9387 7388
www.dallasdelta.com

<<

Page 1

>>

CSV File

Index	Name	Room Number	Phone Number	Relay Code1	Relay Code2	Lift
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						

PC File Storage

Load CSV

Save CSV

Phone Storage

Write

Reload

Exit

Clear All

In this section, fill in the corresponding fields,

The first column '**Name**' will be displayed on the LCD to the user. It should indicate the tenant name or unit number.

The **Room Number** column is a number that can be entered at the phone to call the tenant and can only be numerical digits 0-9, upto 6 digits long.

Note that if you have more than one locations with the same room number, that the first one on the list is used.

The 3rd column is the **Phone Number** is the clients telephone number, any number can be used here If the PBX allows mobile or land line numbers, else they will need to be PBX extensions numbers for each apartment.

The next two columns **Relay Code 1 & 2** will be the relay access codes (4 digits each). These codes are used to in two ways.

Firstly, when in conversation with the phone the tenant can enter the code on there phones keypad to let the visitor in.

Secondly, The tenant can use it as a keyless access at the gate. this is done by pressing '**# roomNumber # code**'.

This event is also sent to Syslog and SNMP servers if enabled, to record for security purposes.

The last column is used in conjunction with a **lift controller** if one is installed, it indicates which level to send the lift. If this is not used then leave this column to 0.

Save the table to computers storage as data is entered by clicking **Save CSV**, and before it can be uploaded to the Phone Storage by clicking **Write**.

Note that ALL data is sorted in alphabetical order (the Name column) prior to Save and Write.



8.0 Release Notes

DATE:	VERSION:	REVISION NOTES:
1/5/19	PCB: S_11	Quick Start Guide and Manual
14/1/20	PCB: S_11_R	Remote relay option
14/5/20	PCB: S_11_Sent	Sentinel phone option

WARNING

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